EXP.NO: 11	Implement an application that writes data to the SD card
DATE:	

AIM:

To develop an Android application using Android Studio that allows users to enter data and store it in the device's SD card (external storage).

ALGORITHM:

1. Input:

- The user enters text data in the application.
- The user clicks a button to save the data into the SD card.

2. Steps for Implementation in Android Studio:

1. Create a New Android Project:

- Open **Android Studio**
- Create a new project with an Empty Activity
- Choose Java or Kotlin as the programming language

2. Design the User Interface:

- Add the following components in the layout:
  - EditText → to enter data
  - Button → to save data
  - TextView (optional) → to display status messages

3. Add Permissions in Manifest File:

- Declare permission to write to external storage
- For modern Android versions, also handle runtime permissions

4. Handle Runtime Permissions:

- Check if permission is granted
- If not, request permission from the user at runtime
- Proceed only after permission is granted

5. Check External Storage Availability:

- Ensure that:
    - SD card is available
    - SD card is writable

6. Create File in SD Card:

- Define a file name (e.g., mydata.txt)
- Select appropriate directory:
    - App-specific folder (recommended)
    - Or public directory

7. Write Data to SD Card:

- Read input from EditText
- Open file output stream
- Write the data into the file
- close the stream

8. Display Output to User:

- Show confirmation message:
    - “Data saved successfully”
    - Or error message if something fails

9. Handle Exceptions:

- Manage:

- IOException
- Permission denial
- Storage unavailability

3. Testing the Application:

- Run the app on a real device/emulator
- Enter sample text
- Click Save button
- Verify file using file manager in external storage

4. Observations:

- Data is stored successfully in the SD card
- Runtime permission handling is essential
- External storage path must be handled carefully in newer Android versions

OUTPUT:

- User enters text → Clicks Save
- Message displayed:

Data saved successfully to SD card

- File mydata.txt created in external storage containing user input

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center"
```

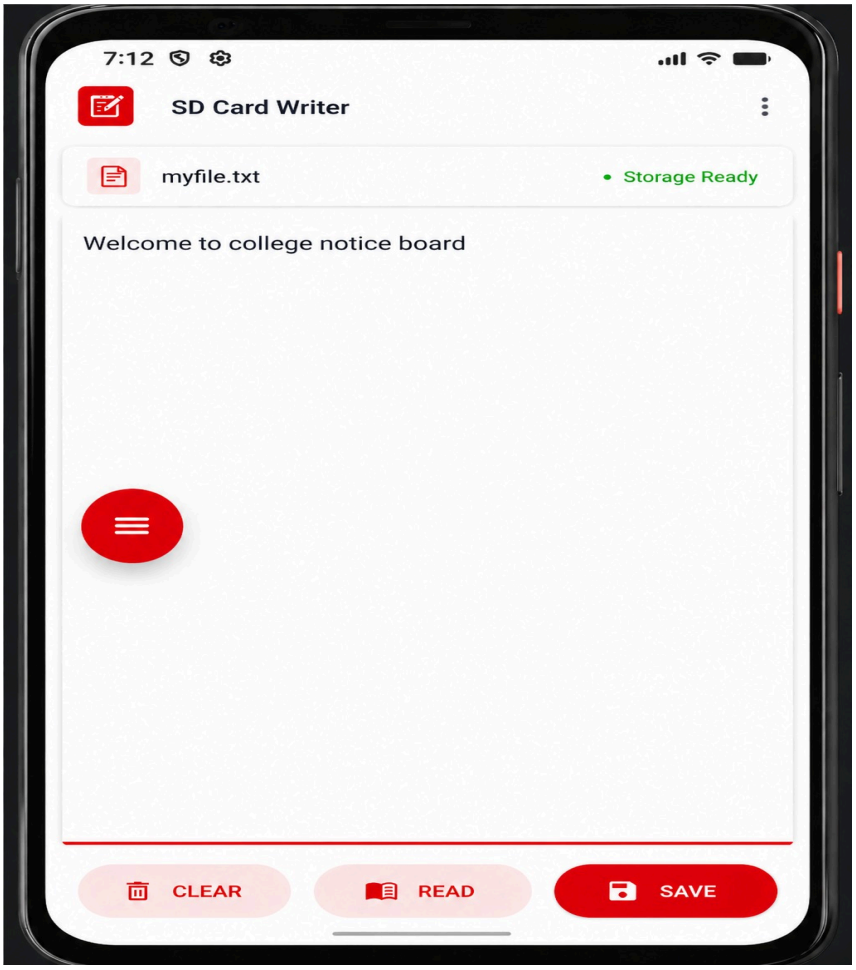
```
    android:background="#F3E5F5">
    <EditText
        android:id="@+id/editTextData"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter text to save"
        android:padding="10dp"
        android:backgroundTint="#9C27B0"/>
    <Button
        android:id="@+id/btnSave"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Save to SD Card"
        android:layout_marginTop="20dp"
        android:backgroundTint="#BA68C8"
        android:textColor="#FFFFFF"/>
</LinearLayout>
```

MainActivity.kt:

```
package com.example.sdcardwrite
import android.os.Bundle
import android.os.Environment
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import java.io.File
import java.io.FileOutputStream
class MainActivity : AppCompatActivity() {
```

```
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)
    val editText = findViewById<EditText>(R.id.editTextData)
    val button = findViewById<Button>(R.id.btnSave)
    button.setOnClickListener {
        val text = editText.text.toString()
        if (text.isEmpty()) {
            Toast.makeText(this, "Enter some text", Toast.LENGTH_SHORT).show()
        } else {
            writeToFile(text)
        }
    }
}   private fun writeToFile(data: String) {
    try {
        val file = File(getExternalFilesDir(null), "myfile.txt")
        val fos = FileOutputStream(file)
        fos.write(data.toByteArray())
        fos.close()
        Toast.makeText(this, "Saved to: ${file.absolutePath}", Toast.LENGTH_LONG).show()
    } catch (e: Exception) {
        e.printStackTrace()
        Toast.makeText(this, "Error saving file", Toast.LENGTH_SHORT).show()
    }}}
```

Output:



RESULT:

An Android application is successfully developed using Android Studio to write data to the SD card with proper permission handling and storage management.